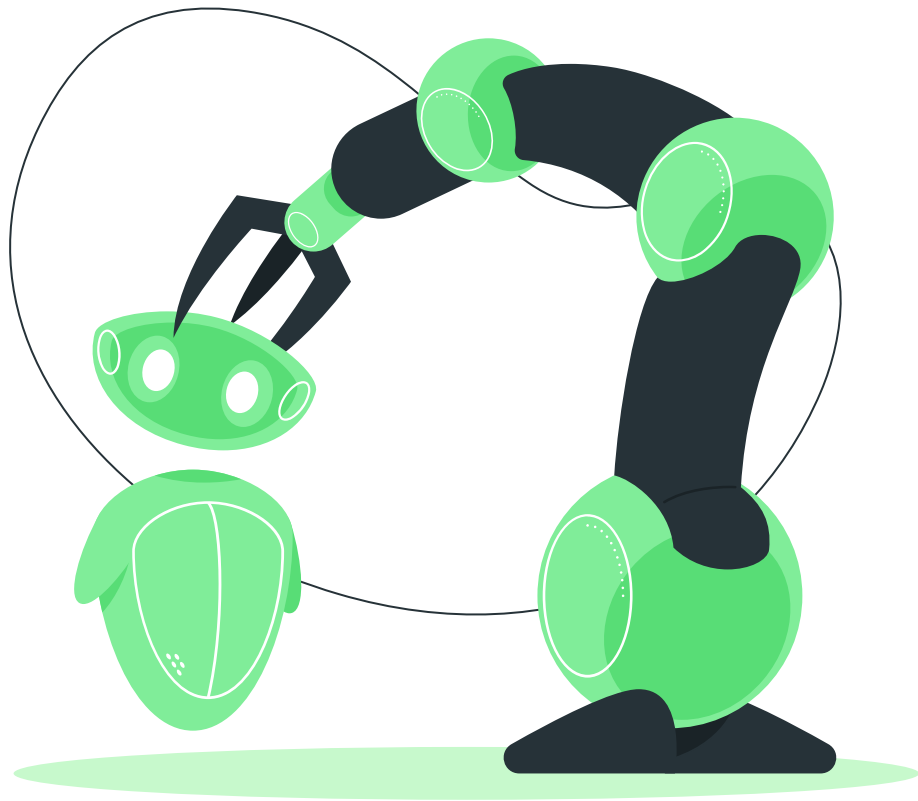


Soutenance finale

Projet n°2: Robot Delta

Grp n°5



Sommaire

1

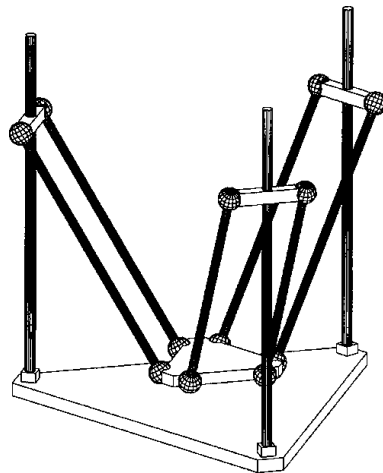
Partie mécanique et simulations

2

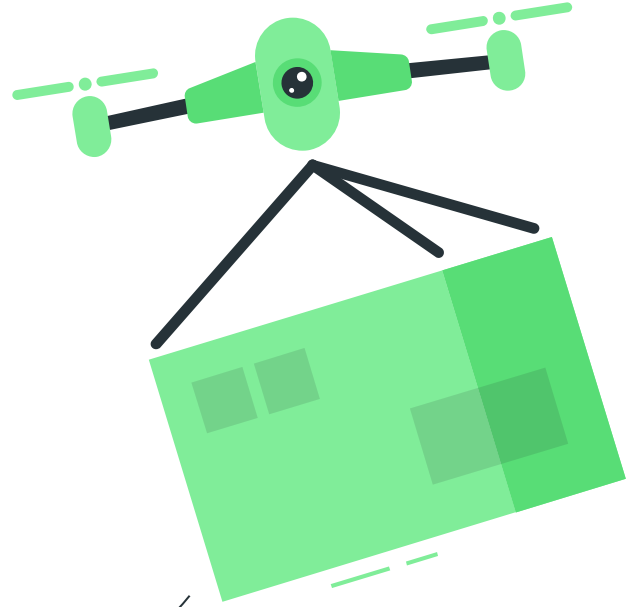
Partie électrique et programmation

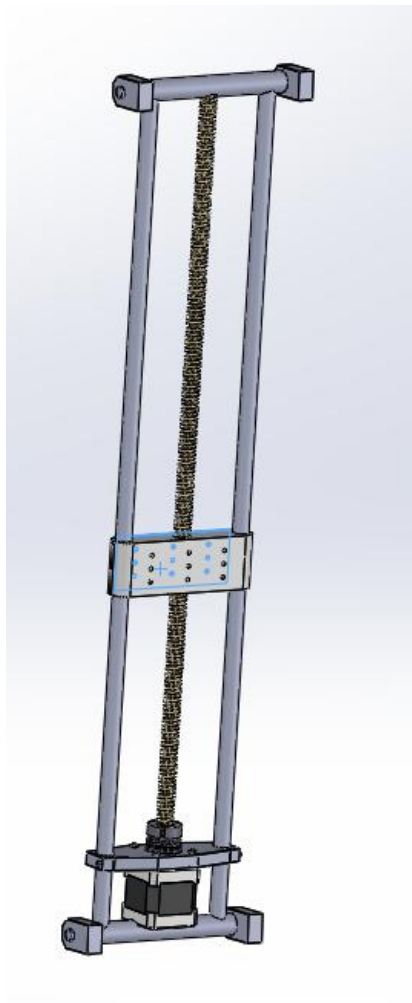
3

Démonstration & Questions



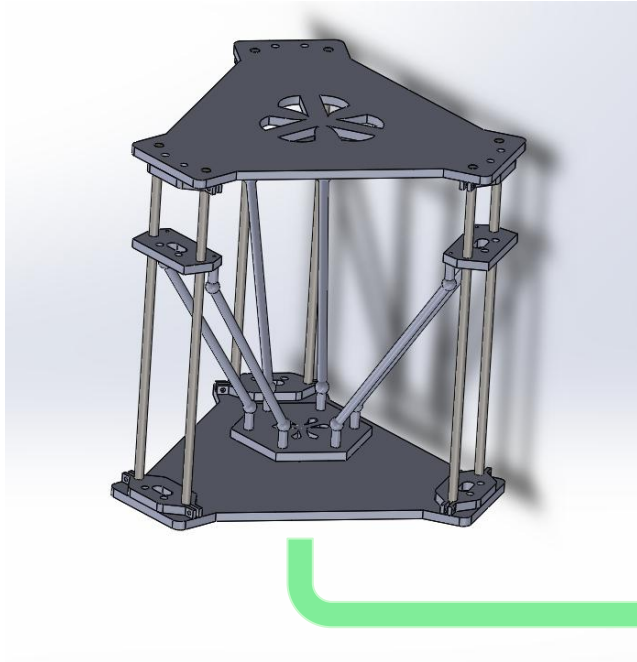
Partie **mécanique**



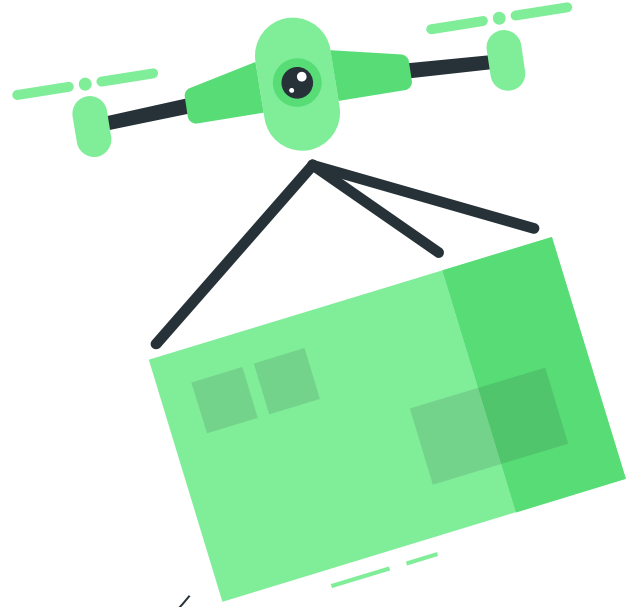


Solution **commune**

Maquette numérique du robot Delta



Partie **électrique**

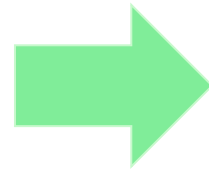




Code C de cinématique inverse

Code C de la cinématique

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <math.h>
4
5 int main()
6 {
7     float x,y,z,r,xc,yc,zc,teta=0,pas,t=0;
8     int n;
9     float z1;
10    float z2;
11    float z3;
12    printf("Donner la valeur du rayon et les coordonées du rayon");
13    scanf("%f %f %f %f",&r,&xc,&yc,&zc);
14    printf("donner le nombre de points");
15    scanf("%d",&n);
16    pas=360/n;
17    z1=zc+sqrt(pow(233,2)-pow((xc-178),2)-pow((yc-75),2));
18    z2=zc+sqrt(pow(233,2)-pow((xc+178),2)-pow((yc-75),2));
19    z3=zc+sqrt(pow(233,2)-pow(xc,2)-pow((yc+150),2));
20    while (teta<360)
21    { teta=teta+pas;
22      x=r*cos(teta*57.29)+xc;
23      y=r*sin(teta*57.29)+yc;
24      //printf("%f %f %f ",r,x,y);
25      z1=zc+sqrt(pow(233,2)-pow((x-178),2)-pow((y-50),2));
26      z2=zc+sqrt(pow(233,2)-pow((x+178),2)-pow((y-50),2));
27      z3=zc+sqrt(pow(233,2)-pow(x,2)-pow((y+150),2));
28      printf("%f,%.2f\n",t,z3);
29      t+=0.5;
30    }
31
32
33    return 0;
34 }
```



```
z2=zc+sqrt(pow(233,2)-pow((x+178),2)-pow((y-50),2));
z3=zc+sqrt(pow(233,2)-pow(x,2)-pow((y+150),2));
printf("%f,%.2f\n",t,z3);
t+=0.5;
```



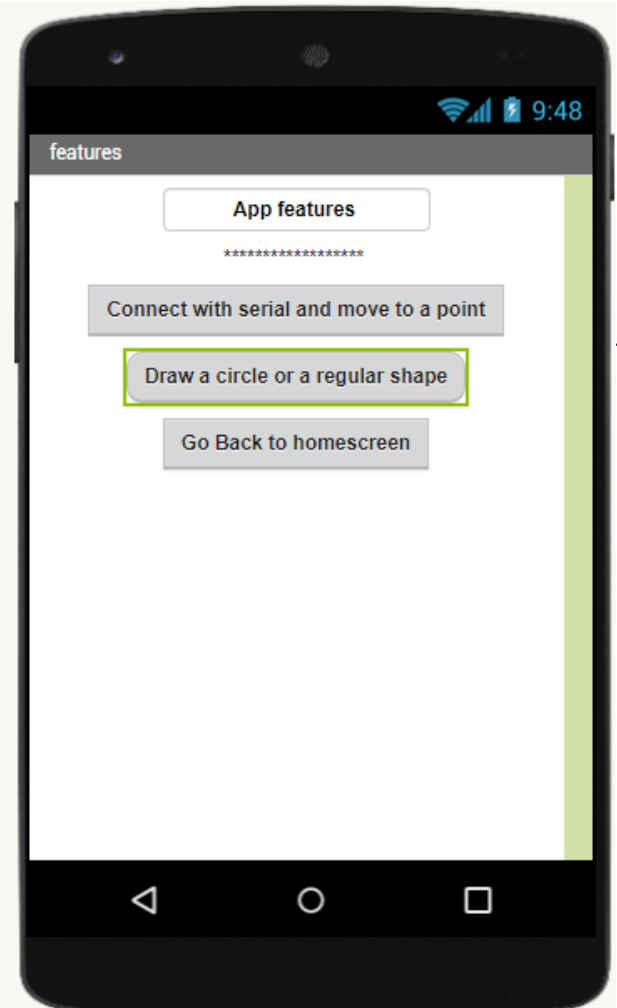
```
Donner la valeur du rayon et les coordonées du rayon20
0
0
0
donner le nombre de points50
0.000000,191.62
0.500000,190.48
1.000000,174.79
1.500000,160.39
2.000000,164.40
2.500000,181.79
3.000000,193.06
3.500000,187.44
4.000000,170.13
4.500000,159.37
5.000000,168.33
5.500000,185.95
6.000000,193.31
6.500000,183.55
7.000000,165.90
7.500000,159.80
8.000000,172.86
8.500000,189.36
9.000000,192.35
9.500000,179.08
10.000000,162.50
10.500000,161.65
11.000000,177.61
11.500000,191.81
12.000000,190.24
12.500000,174.35
13.000000,160.24
13.500000,164.73
```



1

Application de commande

Interface graphique



Circle & Regular Shapes

Enter the radius

Enter the x coordinate of the center

Enter the y coordinate of the center

Enter the z coordinate of the center

Proceed

Enter the number of points

to draw a circle choose a number higher than 36
3 will result in a triangle, 4 in a square, 5 a pentagone...

Go back

Les commandes

Serial

Establishing the bluetooth connection

connect disconnect

☐ connected

Go back to App features

Inserting Coordinates

x Coordinate

y Coordinate

z Coordinate

Proceed



Code

Arduino

Vue du code

Code_final_delta_laila

```
int pulsex=6, pulsey=9, pulsez=12;
int dirx=5 , diry=8 , dirz=11;
int enablex=4, enabley=7, enablez=10;
float x,y,z;
char commande;
String inststring;
String xcl,ycl,zcl,rl,xli,yli,xl,yl,zl;
float npulsionx,npulsiony,npulsionz;
float pulsion1,pulsion2,pulsion3;
float m;
int tb=10;
int a,b,d;
int th=100;
const byte interruptX=19;
const byte interruptY=20;
const byte interruptZ=18;
String message="";
```

Conclusion

